

Bluestock Mutual Fund Analytics Capstone

Final Project Report

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Technology Stack: Python | SQL | Power BI | Data Analytics

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1. Executive Summary

The Bluestock Mutual Fund Analytics Capstone project focuses on analyzing Indian mutual fund data to understand investment trends, fund performance, asset growth, and investor behavior.

The project implements an end-to-end analytics workflow including data extraction, data cleaning, exploratory data analysis, performance analysis, and interactive dashboard development.

A Python-based ETL pipeline was developed to automate data preparation. The processed datasets were analyzed using Python and visualized through Power BI dashboards.

The project provides insights into:

- Assets Under Management (AUM) growth
- NAV trends
- SIP investment patterns
- Fund performance comparison
- Category-wise investment behavior

2. Problem Statement

Mutual fund data is available across multiple datasets, making it difficult to analyze investment trends and compare fund performance.

The objective of this project is to build an analytics solution that combines multiple datasets and provides meaningful insights through dashboards.

3. Project Objectives

The main objectives are:

- Analyze mutual fund scheme performance
- Understand AUM growth patterns
- Study SIP inflow trends
- Compare fund categories
- Identify top-performing funds

- Build an automated ETL pipeline
- Create interactive Power BI dashboards

4. Data Sources

The project uses multiple mutual fund datasets to perform comprehensive analysis.

Dataset	Description
Fund Master	Contains mutual fund scheme details such as scheme name, category, and fund house
NAV History	Contains historical Net Asset Value records
AUM Data	Provides Assets Under Management information
SIP Inflows	Contains monthly SIP investment trends
Category Inflows	Represents investment flows across mutual fund categories
Folio Count	Provides investor participation information
Scheme Performance	Contains fund return and performance metrics
Investor Transactions	Contains transaction-level information

These datasets were combined to create a complete view of mutual fund performance and investor behavior.

5. ETL Architecture

The project follows an automated ETL (Extract, Transform, Load) workflow.

Extract

The extraction process reads mutual fund CSV datasets from the processed data directory.

Transform

The transformation stage performs:

- Duplicate removal
- Column name standardization
- Missing value handling
- Data formatting
- Data preparation for analysis

Load

The cleaned datasets are stored for further analysis and dashboard development.

ETL Flow

Raw Dataset Files



Extract (Python)



Transform (Cleaning)



Load (Processed Data)



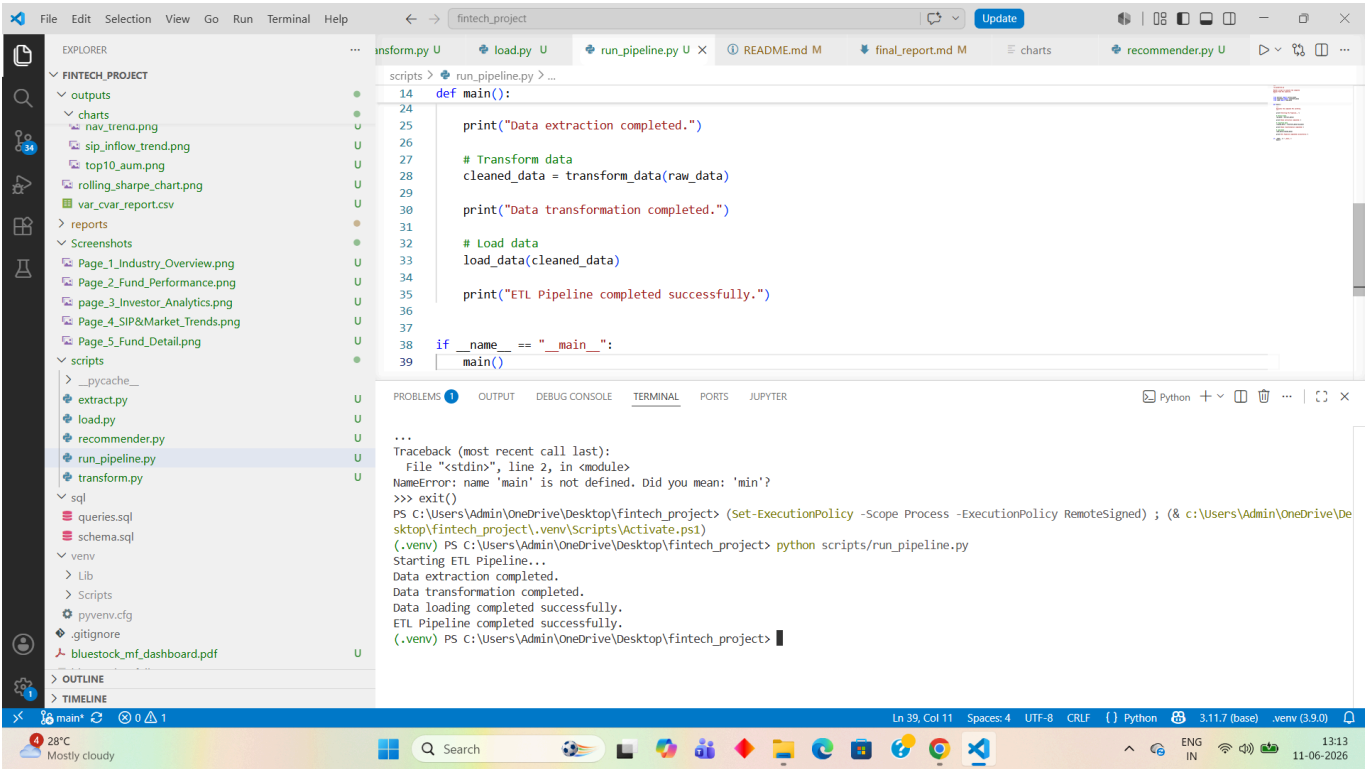
Analytics Dashboard

The pipeline is automated using:python scripts/run_pipeline.py

ETL Pipeline Implementation

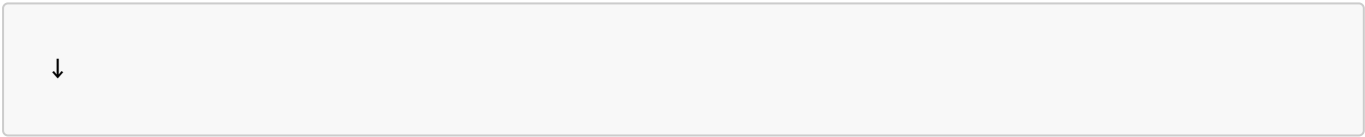
The ETL pipeline was implemented using modular Python scripts:

- extract.py → Reads raw datasets
- transform.py → Cleans and prepares data
- load.py → Stores processed outputs
- run_pipeline.py → Executes complete workflow

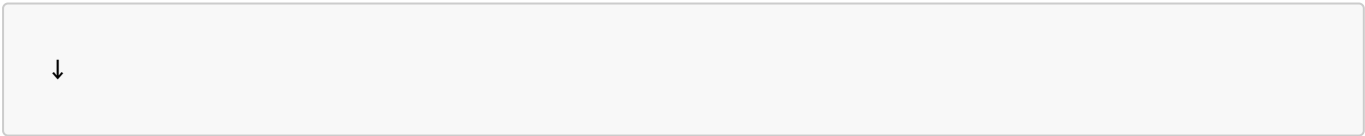


System Architecture

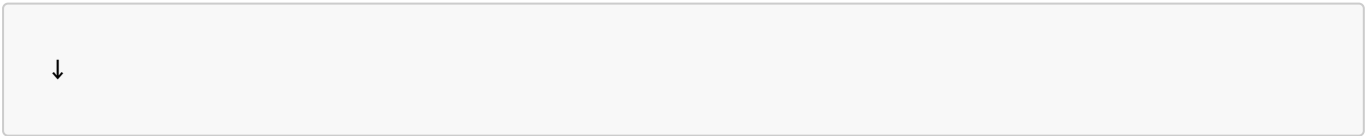
The overall architecture of the project is: Raw Mutual Fund Data



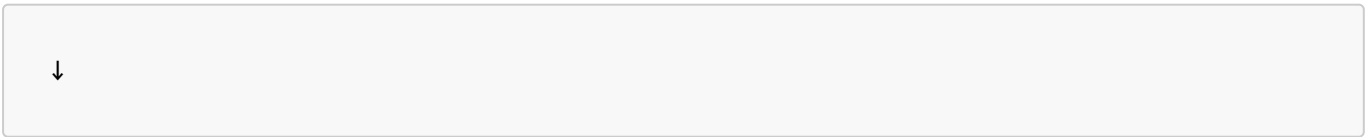
Python ETL Pipeline



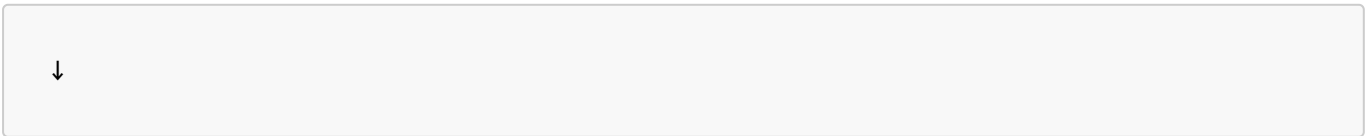
Clean Processed Data



EDA & Performance Analysis



Power BI Dashboard



Business Insights

6. Exploratory Data Analysis

EDA was performed to understand data distribution, trends, and relationships.

AUM Analysis

The Assets Under Management analysis identifies growth patterns across mutual fund categories.

Key observations:

- Certain categories contribute significantly to total AUM.
- Large fund houses dominate the market share.
- AUM trends help understand investor confidence.

NAV Trend Analysis

Historical NAV analysis was performed to study fund value movement over time.

Insights:

- NAV values show variation across schemes.
- Long-term trends help compare fund performance.

SIP Inflow Analysis

SIP inflow analysis provides insights into investor participation.

Findings:

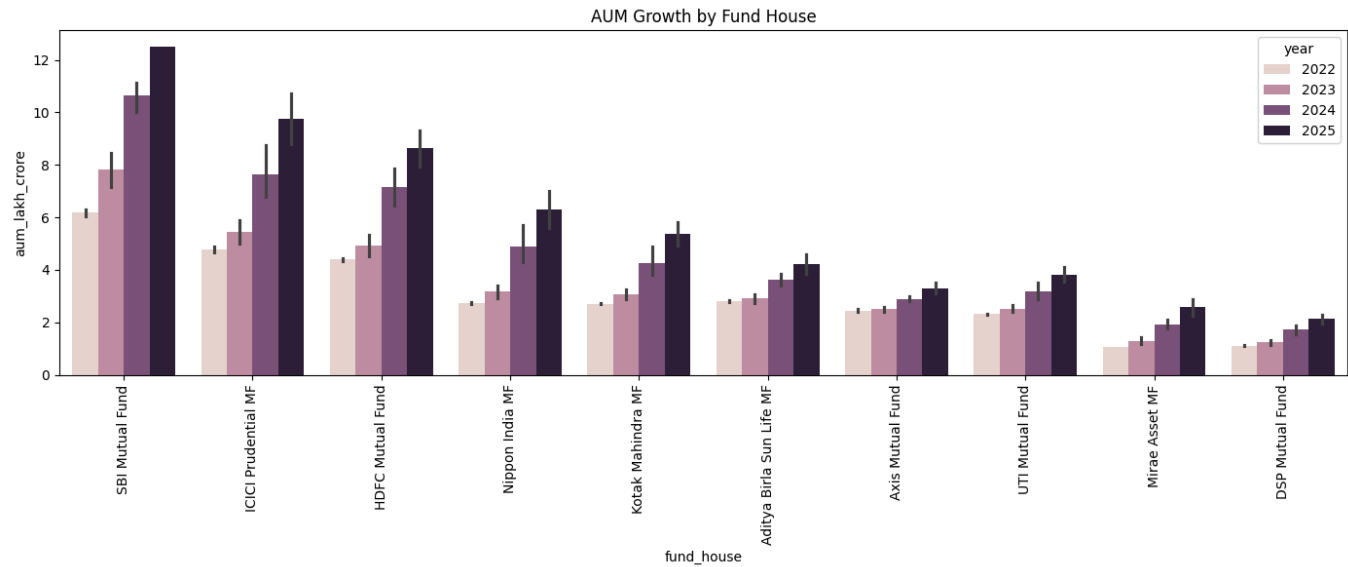
- SIP investments show increasing adoption.
- Regular investment patterns indicate long-term investor interest.

6.1 EDA Visualizations

AUM Growth Analysis

The AUM growth analysis shows the change in Assets Under Management over time.

AUM is an important indicator of investor participation and overall mutual fund market growth.

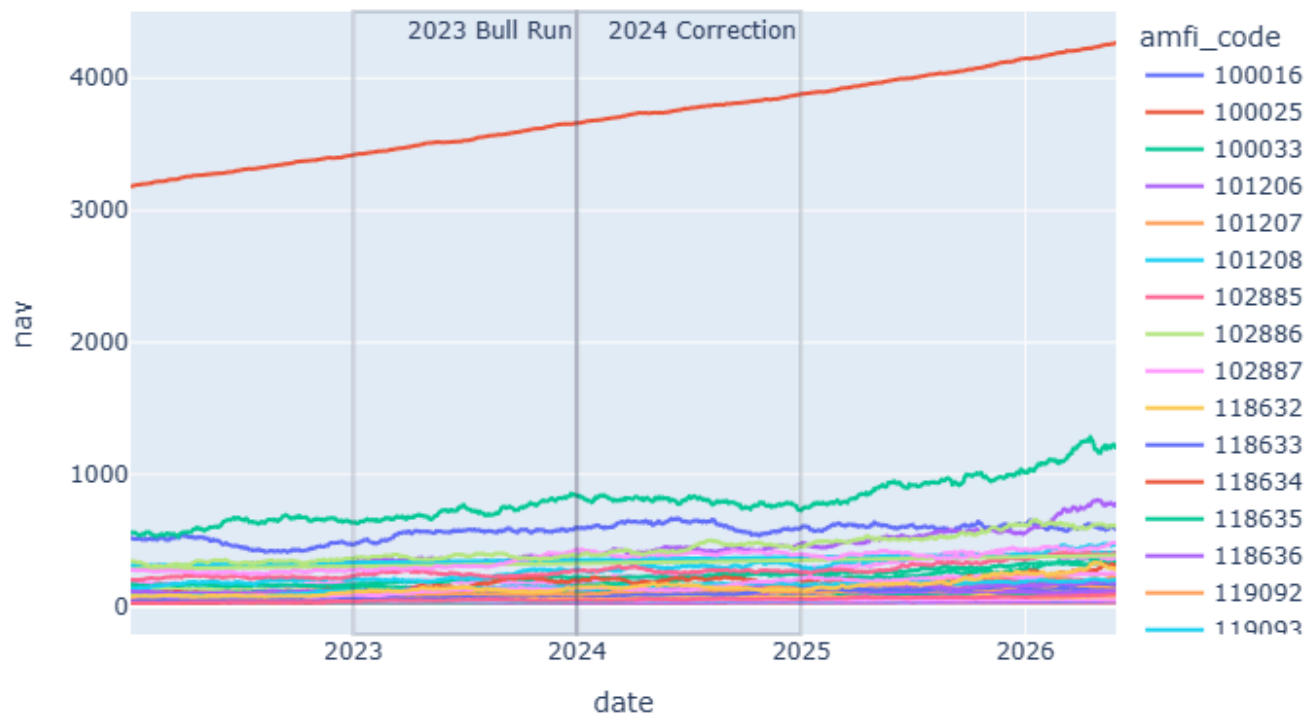


NAV Trend Analysis

The NAV trend analysis represents the historical movement of mutual fund Net Asset Value.

This helps understand fund value changes and long-term performance patterns.

Daily NAV Trend (2022-2026)



SIP Inflow Analysis

The SIP inflow analysis highlights monthly investment patterns.

The trend helps identify investor participation and systematic investment behavior.

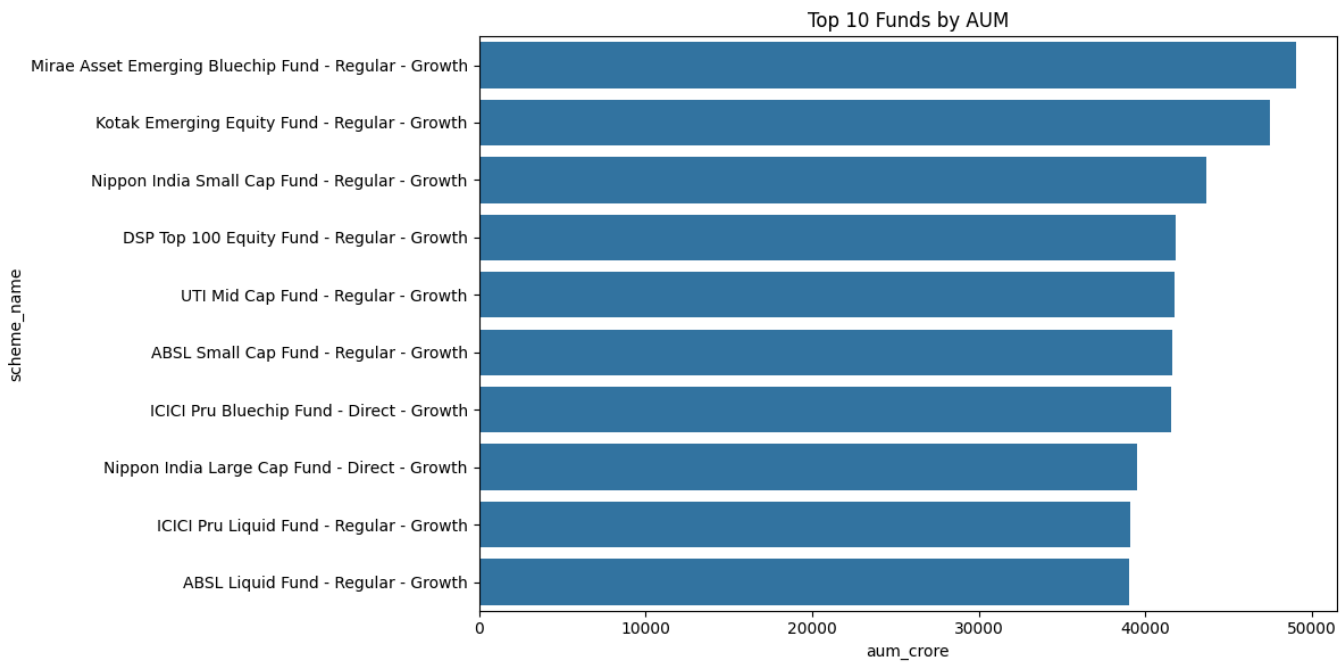
Monthly SIP Inflows (2022-2025)



Top Funds by AUM

This analysis identifies mutual fund schemes with the highest Assets Under Management.

Higher AUM generally indicates greater investor trust and market presence.



7. Performance Analysis

Performance analysis was performed to evaluate mutual fund schemes based on returns, growth trends, and investment patterns.

Fund Performance Evaluation

The analysis compares different mutual fund schemes using available performance metrics.

The evaluation focuses on:

- Historical returns
- NAV movement
- Fund category performance
- Assets Under Management (AUM)
- Investor participation

Performance Insights

The analysis identified differences in performance across various mutual fund categories.

Key observations:

- Some categories demonstrate stronger growth potential.
- Large AUM funds show higher investor participation.
- Historical performance helps compare scheme effectiveness.

Category Performance Analysis

Category-level analysis helps understand where investors are allocating capital.

It provides insights into:

- Equity fund growth
- Debt fund behavior
- Hybrid fund trends

SIP Performance Analysis

SIP analysis evaluates systematic investment behavior.

Findings:

- SIP investments represent consistent investor participation.
- Monthly trends help identify investment patterns.
- Increasing SIP flows indicate growing awareness of mutual fund investments.

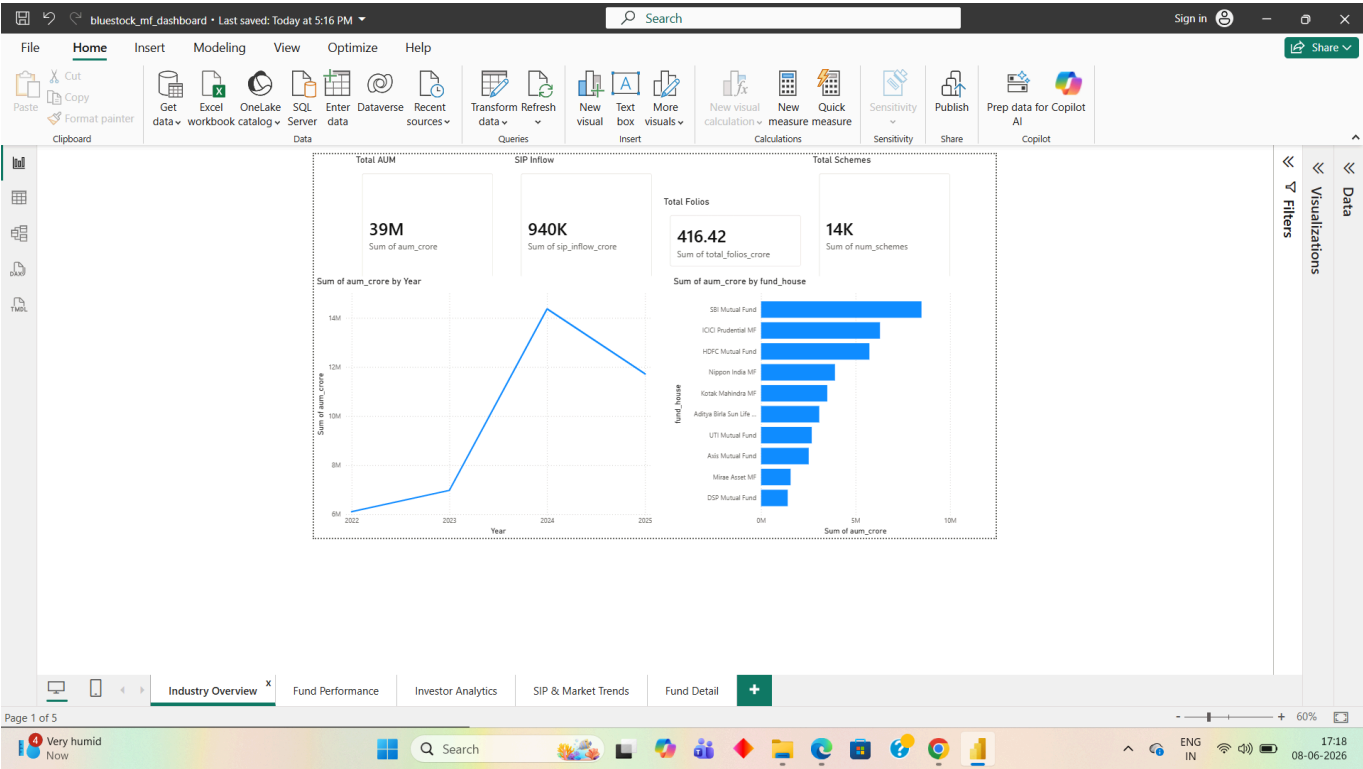
8. Dashboard Screenshots

The Power BI dashboard was developed to provide interactive insights into mutual fund performance, investor behavior, SIP trends, and fund-level analysis.

The dashboard consists of multiple pages covering different analytical views.

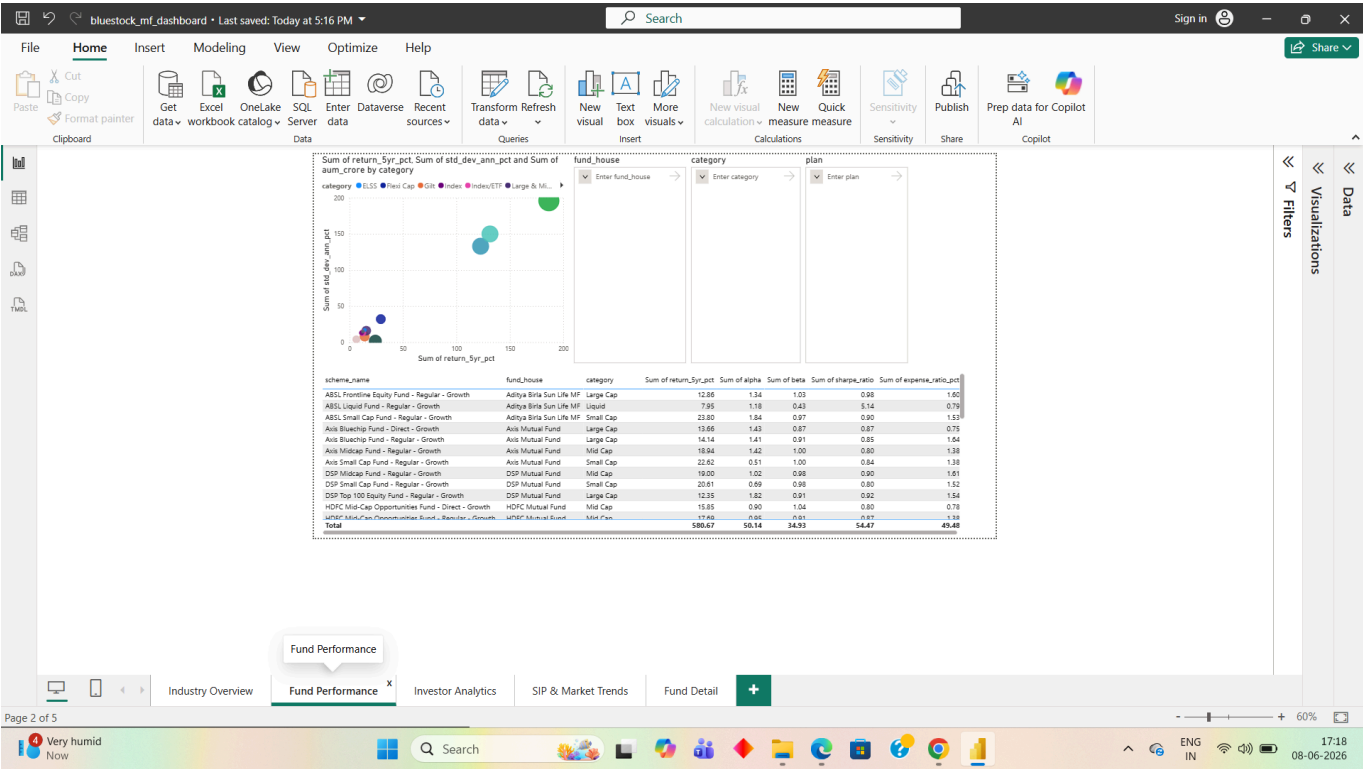
Page 1: Industry Overview

This dashboard page provides a high-level overview of the mutual fund industry including major trends and summary metrics.



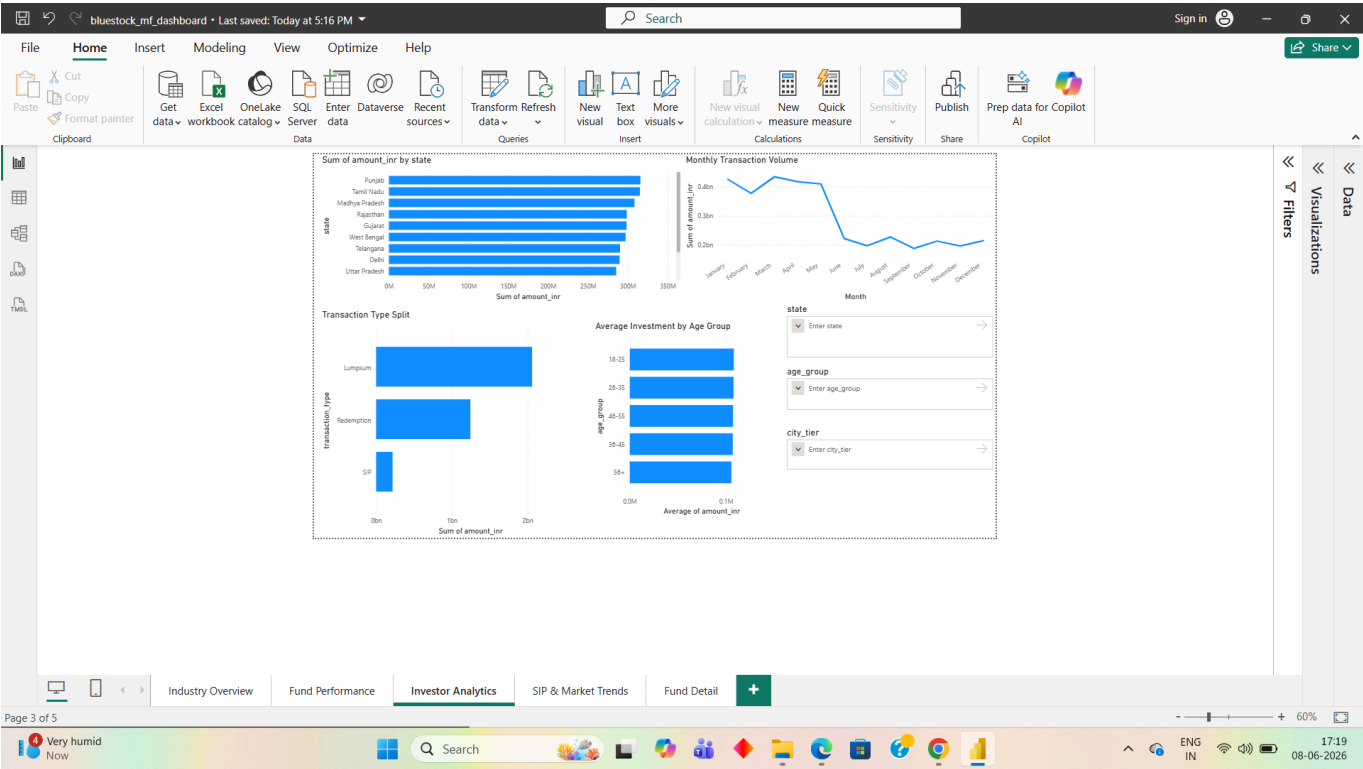
Page 2: Fund Performance

This page focuses on comparing mutual fund performance and analyzing fund-level metrics.



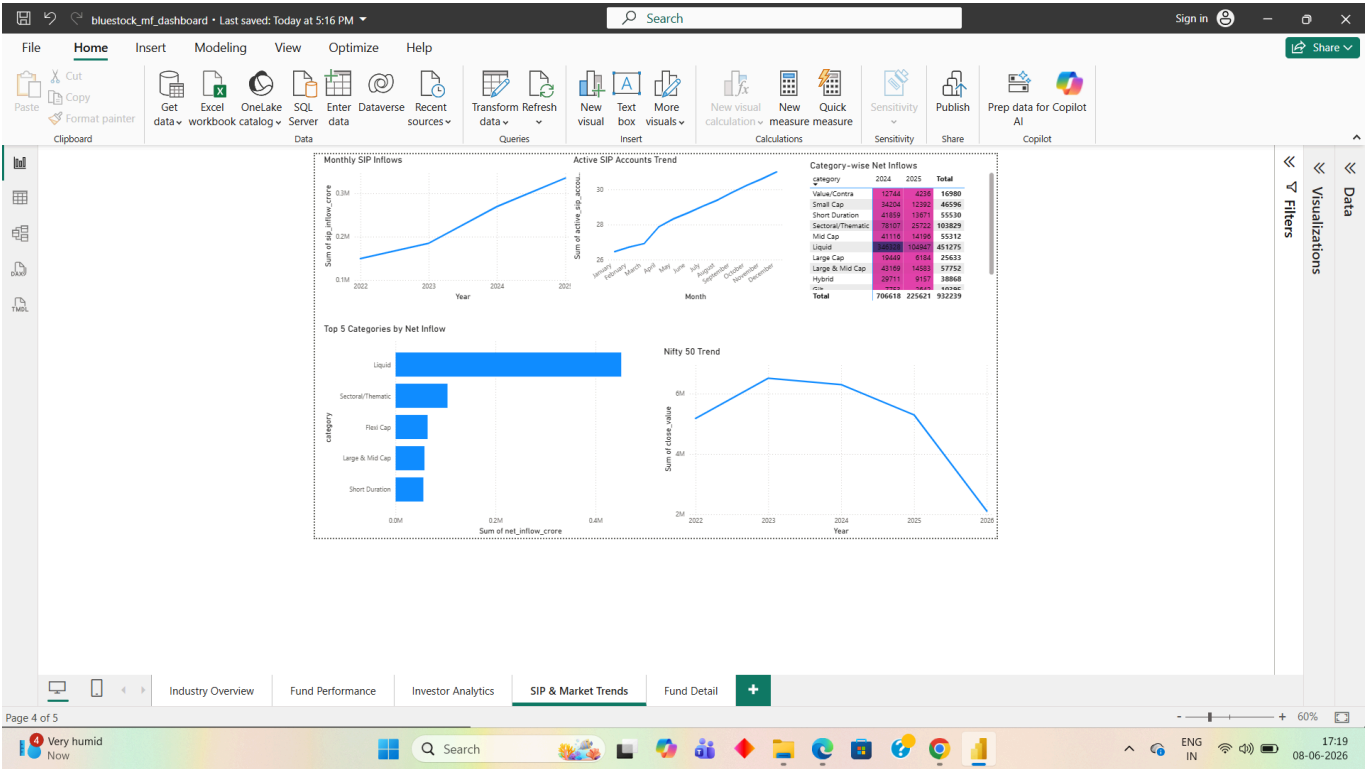
Page 3: Investor Analytics

This dashboard page analyzes investor-related patterns and participation trends.



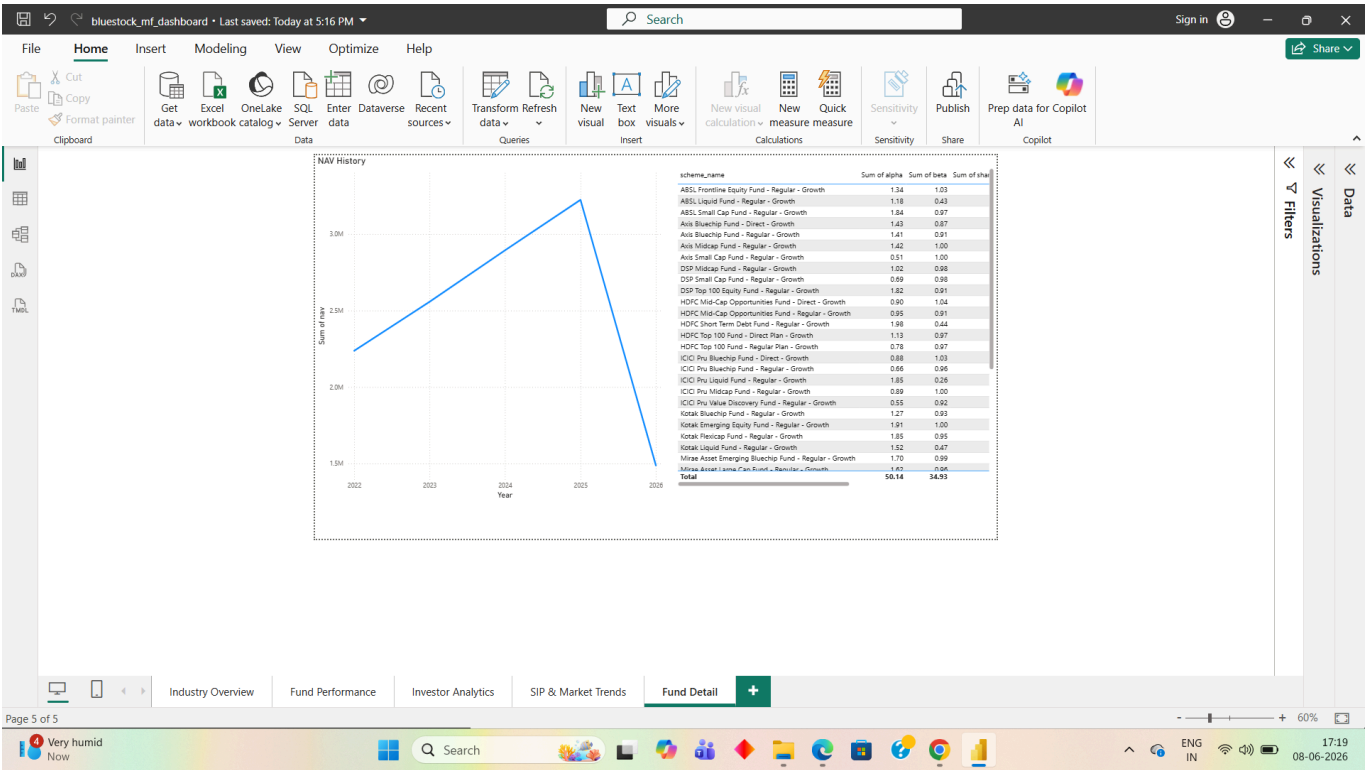
Page 4: SIP & Market Trends

This page highlights systematic investment plan trends and market movement insights.



Page 5: Fund Details

This page provides detailed information about individual mutual fund schemes.



9. Limitations

Although the project provides meaningful insights into mutual fund analytics, there are certain limitations.

Data Availability

The analysis depends on the datasets available for the project. Limited historical records may affect long-term trend analysis.

External Market Factors

The project does not include external factors such as:

- Market volatility
- Economic conditions
- Interest rate changes
- Government policy changes

Real-Time Data

The dashboard uses static datasets and does not currently support live market updates.

Predictive Analysis

The current project focuses on descriptive and diagnostic analytics. Future versions can include predictive models for forecasting fund performance.

10. Recommendations

Based on the analysis performed, the following recommendations can improve the solution:

Real-Time Data Integration

Integrate financial APIs to automatically update:

- NAV values
- Market prices
- Fund performance metrics

Predictive Analytics

Machine learning models can be implemented to predict:

- Future fund performance
- Investor behavior
- Market trends

Risk Analysis Enhancement

Include additional risk metrics:

- Volatility
- Sharpe Ratio
- Benchmark comparison

- Maximum drawdown

Dashboard Improvements

Future dashboard versions can include:

- Automated refresh
- User-based filtering
- Personalized fund recommendations

11. Future Scope

The project can be enhanced with advanced analytics capabilities.

Future improvements include:

- Real-time mutual fund tracking
- AI-based investment recommendations
- Portfolio optimization models
- Risk prediction models
- Automated reporting systems
- Cloud deployment of analytics pipeline

12. Conclusion

The Bluestock Mutual Fund Analytics Capstone successfully developed an end-to-end analytics solution for understanding mutual fund performance and investor trends.

The project combined:

- Data extraction
- Data transformation
- Exploratory analysis
- Performance evaluation
- Dashboard development

The automated ETL pipeline improved data preparation efficiency, while Power BI dashboards provided interactive insights into mutual fund trends.

The analysis helps investors and analysts understand fund behavior, category growth, and investment patterns.

Overall, the project demonstrates the application of data analytics techniques in the financial domain.

Data Quality Assessment

Data quality checks were performed before analysis.

The following checks were applied:

Check	Description
Missing Values	Identified and handled incomplete records
Duplicate Records	Removed duplicate entries
Column Standardization	Unified column naming format
Data Type Validation	Verified correct data formats
Consistency Checks	Ensured reliable analysis results

These quality checks improved the accuracy and reliability of the analytics pipeline.

Project Information

Field	Details
Project Name	Bluestock Mutual Fund Analytics Capstone
Domain	Financial Data Analytics
Tools	Python, SQL, Power BI
Report Prepared By	Shobhitha H R
Date	11 June 2026

13. Appendix

Tools Used

Tool	Purpose
Python	Data processing and analysis
Pandas	Data manipulation
NumPy	Numerical operations
Matplotlib	Data visualization
SQL	Data querying
Power BI	Dashboard development
GitHub	Version control

ETL Execution Command

```
python scripts/run_pipeline.py
```

Project Repository

GitHub repository link will be added after final push.